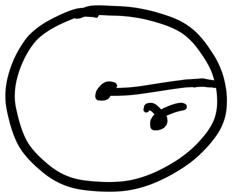
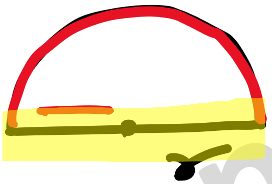
Circle

① 

→ Circle → Perimeter = Circumference $= 2\pi r$ → Area $= \pi r^2$

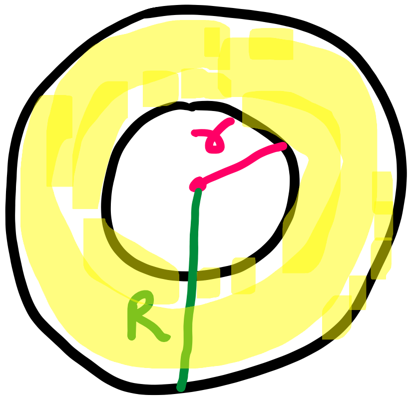
② 

→ Semi-circle → $P = \pi r$ ✗ [wrong] → $A = \frac{\pi r^2}{2}$ //

$\therefore P = \pi r + d$

$= \pi r + 2r$

③



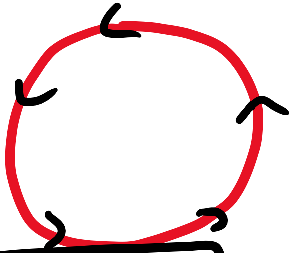
→ Ring shape (annular ring)

→ $P = 2\pi r + 2\pi R$ //

→ $A = A(o) - A(\text{Inner})$

$= \pi R^2 - \pi r^2$ //

Area

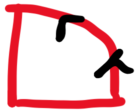


$$\boxed{A = \pi r^2}$$

$$\rightarrow A = \frac{360^\circ}{360^\circ}\pi r^2$$

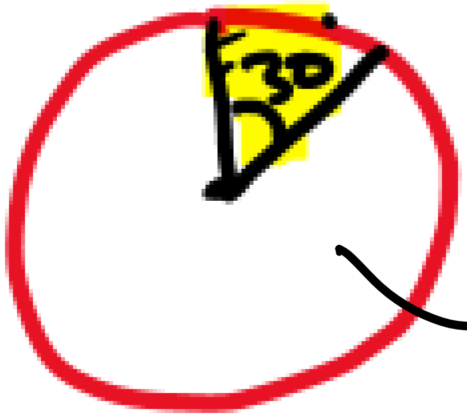
$$\boxed{A = \frac{\pi r^2}{2}}$$

$$\rightarrow A = \frac{180}{360}\pi r^2$$



$$\boxed{A = \frac{1}{4}\pi r^2}$$

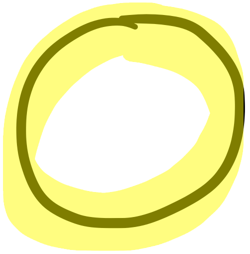
$$\rightarrow A = \frac{90}{360}\pi r^2$$

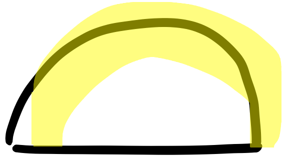


$$A = \frac{30}{360}\pi r^2$$

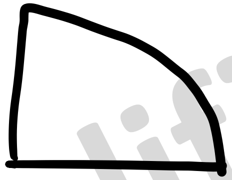
$$\Rightarrow A = \frac{\theta}{360} \times \pi r^2$$

#



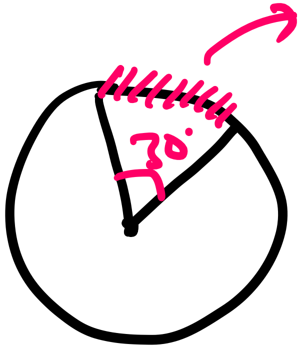


(πr)



$\left(\frac{\pi r}{2}\right)$

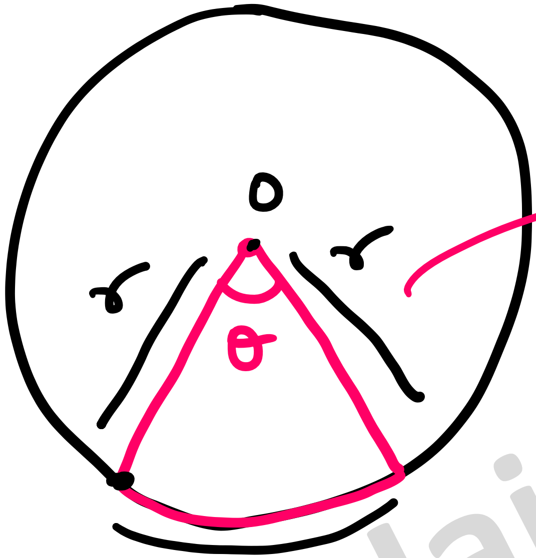
Length of arc = $2\pi r$



$L = \frac{30}{360} \cdot 2\pi r$

$$\boxed{\text{Length of arc} = \frac{\theta}{360} \times 2\pi r}$$

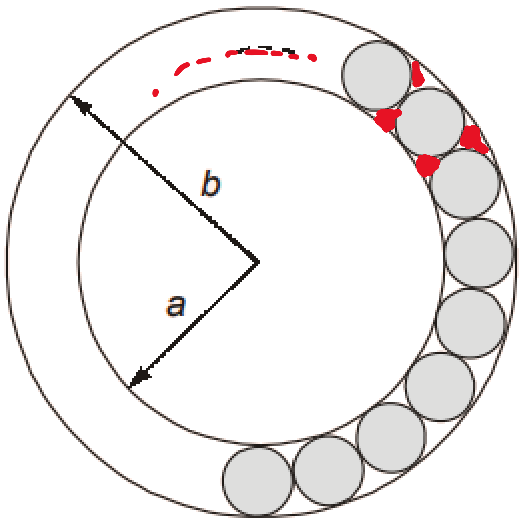
· **Sector**



$$P = r + r + l$$

$$\rightarrow l = \left(\frac{\theta}{360} \cdot 2\pi r\right)$$

The figure below shows an annular ring with outer and inner radii as b and a, respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted are available in annular space is ________.



(a) $\pi[(b^2 - a^2) + n(b - a)^2]$

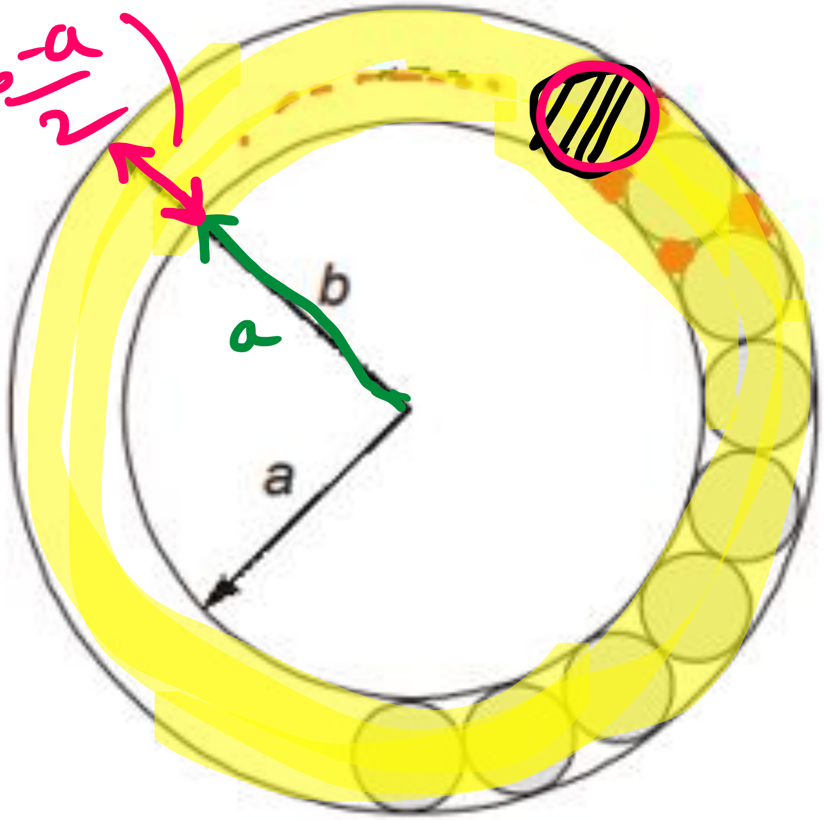
(b) $\pi\left[(b^2 - a^2) + \frac{n}{4}(b - a)^2\right]$

(c) $\pi[(b^2 - a^2) - n(b - a)^2]$

(d) $\pi\left[(b^2 - a^2) - \frac{n}{4}(b - a)^2\right]$

$\therefore d = b-a$

$\text{radius} = \frac{b-a}{2}$



$A(1\,\text{ball}) = \pi r^2$

$= \pi \left(\frac{b-a}{2}\right)^2$

① Area of annular space

$= A(\text{out}) - A(\text{in})$

$= \pi b^2 - \pi a^2$

② Area of unpainted

$= A(\text{annular space})$

$- A(n\ \text{balls})$

$\Rightarrow$ Area of unpainted

= A(annular space)

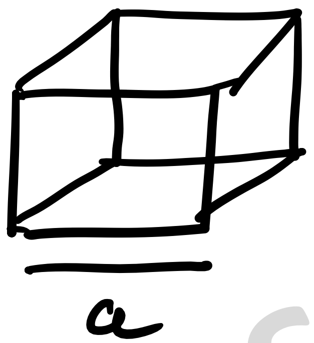
− A(n balls)

$$= \left[\pi b^2 - \pi a^2\right] - n\pi\left(\frac{b-a}{2}\right)^2$$

$$= \pi\left[(b^2-a^2) - \frac{n}{4}(b-a)^2\right]$$

3D Figure

① Cube →



→ Volume = a^3

→ Curved Surface Area

or

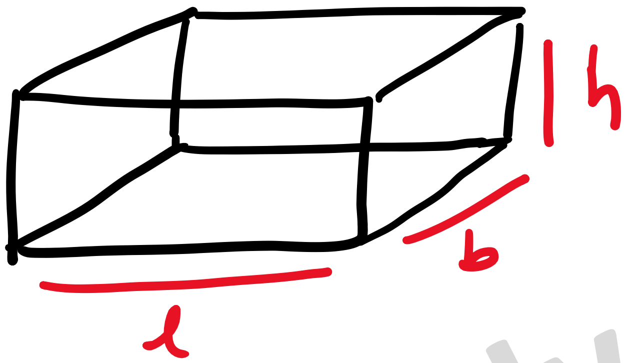
Lateral Surface Area

$= 4a^2$

→ Total Surface Area

$= 6a^2$

② Cuboid →

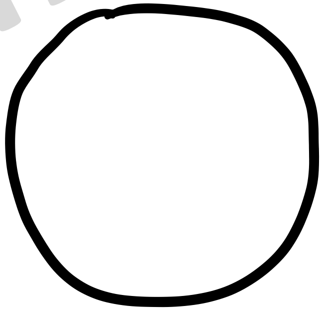


→ $V = lbh$

→ CSA = LSA = $2bh + 2lh$

→ TSA = $2lb + 2bh + 2lh$

③ Sphere →

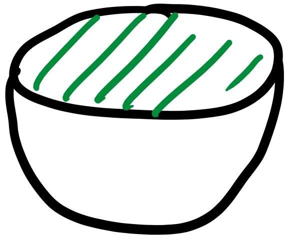


→ $V = \frac{4}{3}\pi r^3$

→ CSA = $4\pi r^2$

→ TSA = $4\pi r^2$

(4) Hemisphere →

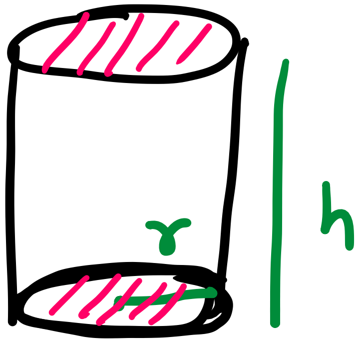


$$V = (2/3)\pi r^3$$

$$CSA = 2\pi r^2$$

$$TSA = 2\pi r^2 + \pi r^2 = 3\pi r^2$$

⑤ Cylinder →



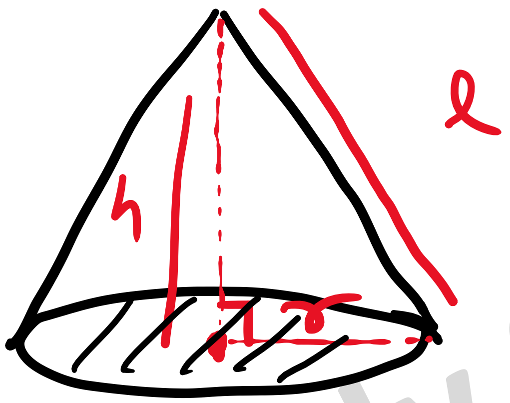
$V = \pi r^2 h$

$CSA = 2\pi r h$

$TSA = 2\pi r h + \pi r^2 + \pi r^2$

$= 2\pi r h + 2\pi r^2$ //

∴ ⑥ Cone →



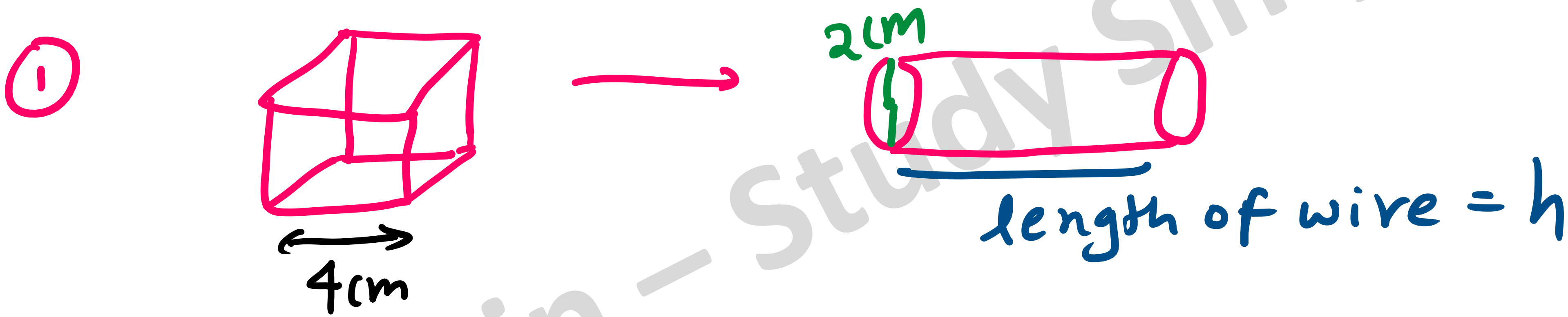
$V = \frac{1}{3}\pi r^2 h$

$LSA = CSA = \pi r l$

$TSA = \pi r l + \pi r^2$

$= \pi r [l + r]$

Q] A steel cube of side 4 cm is melted and a wire of diameter 2 cm is made out of it. Find the length of the wire.



Volume of cube = 4^3

= volume of wire

volume of wire $= \pi r^2 h$

$4^3 = \pi (1)^2 \cdot h$

$\therefore h = \frac{4^3}{\pi}$

$h \approxeq 20.36$ cm

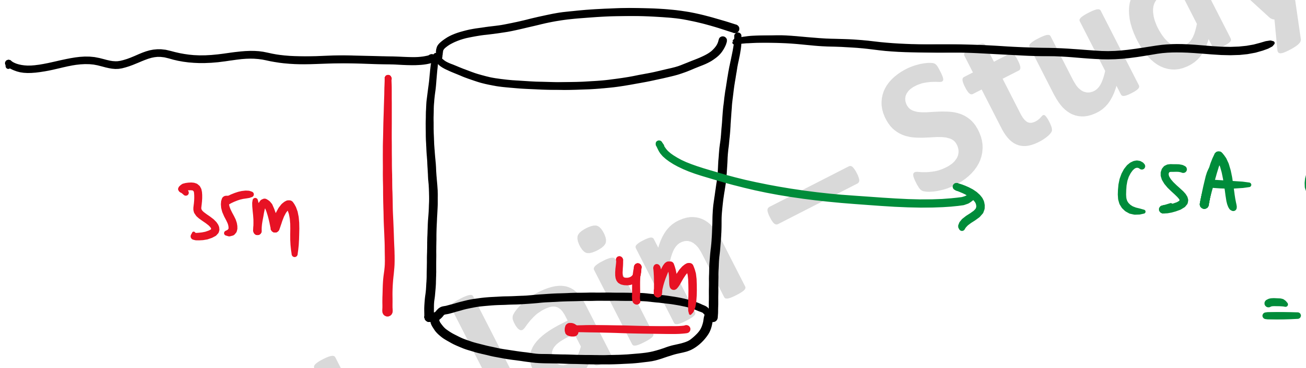
$d = 2$ cm

$2r = 2$ cm

$r = 1$

$\pi = \frac{22}{7}$

Q] A well of radius 4m and height 35 m has been dug out. What is the cost of plastering the inner surface of the well at the rate of Rs 0.8 per sq m?



CSA of cylinder

$= 2\pi rh$

$= 2\pi(4)(35)$

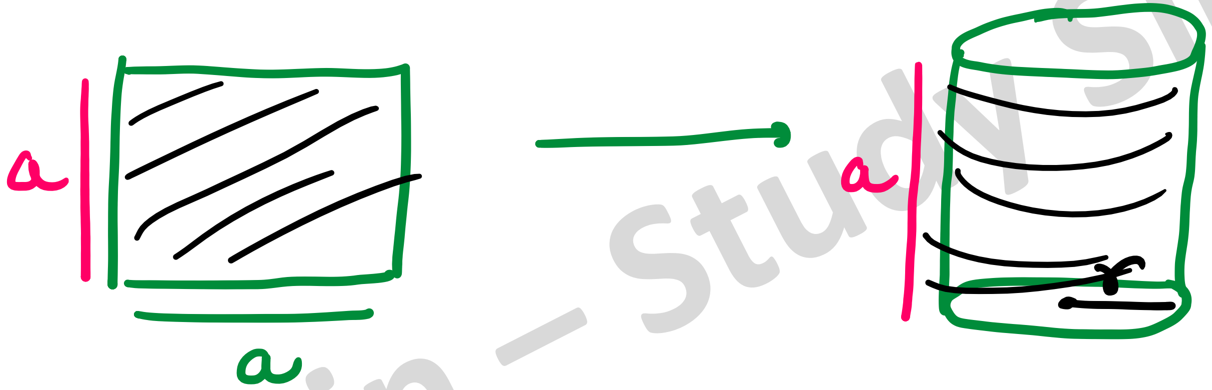
$= 880 m^2$

$1 m^2 \longrightarrow$ Rs 0.8

$880 m^2 \longrightarrow$ Rs $0.8 \times 880 = 704$

Q] A square sheet of paper is converted into a cylinder by rolling it along its length. What is the ratio of base radius to the side of the square?

1



2 Ratio of base radius to side of square

$$= \frac{r}{a} = \frac{a/2\pi}{a} = 1/2\pi$$ //

③ CSA of cylinder

=

Area of square

$\rightarrow \quad 2\pi r h = a^2$

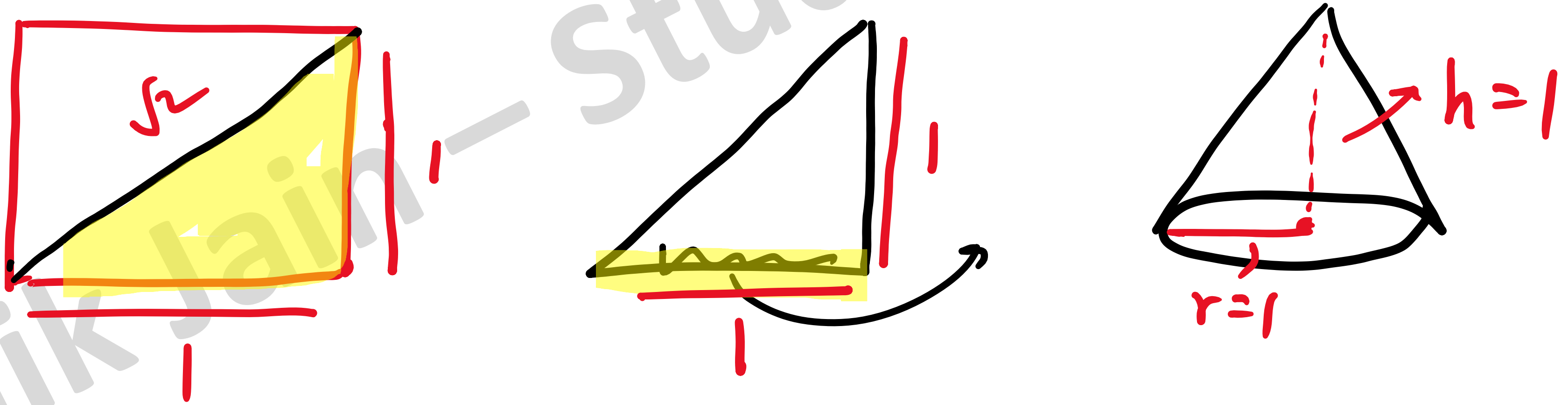
$\rightarrow \quad 2\pi r (\cancel{a}) = a^{\cancel{2}}$

$\rightarrow \quad \boxed{r = \frac{a}{2\pi}}$

Q] Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step, one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in cubic units is _______

[2021]

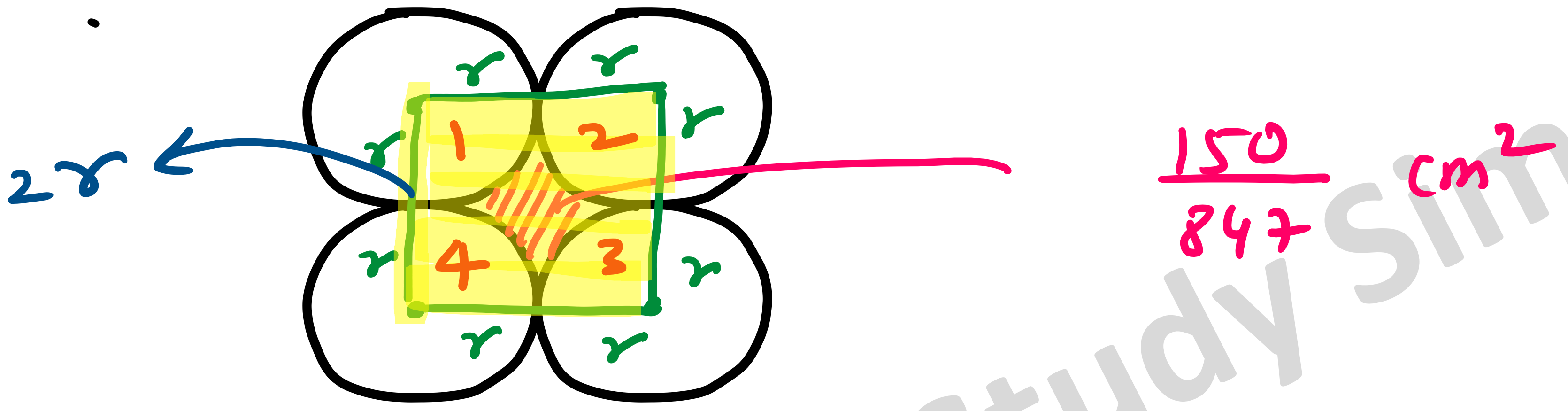
1



2 $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(1)^2(1) = \frac{\pi}{3}$ //

Four equal discs are placed such that each one touches two others. If the area of empty space enclosed by them is 150/847 sq cm, then the radius of each disc is equal to

(a) 7/6 cm
(b) 5/6 cm
(c) 1/2 cm
(d) 5/11 cm



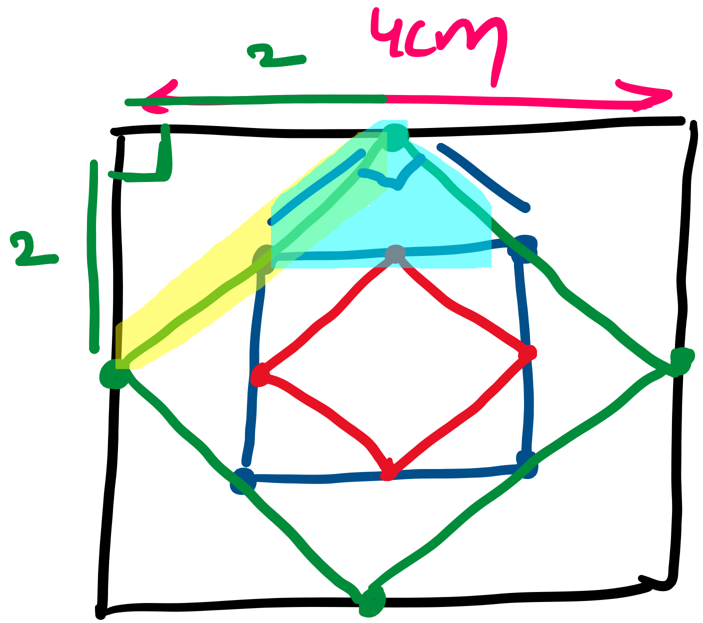
$$A(\text{Empty space}) = A(\square) - A(\circ)$$

$$\frac{150}{847} = (2r)^2 - \pi r^2$$

$$\Rightarrow \quad r = 5/\pi \text{ cm}$$

A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 4 cm, determine the sum of the areas of all the squares :

(a) $32\ cm^2$
(b) $16\ cm^2$
(c) 20
(d) None of these



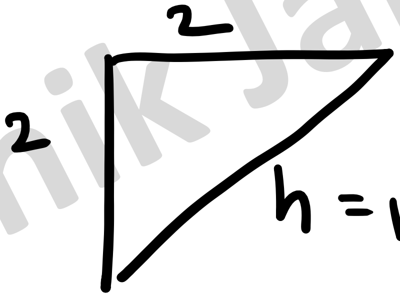
$A(1^{st}\,\square) = 4^2 = 16$

$A(2^{nd}\,\square) = (\sqrt{8})^2 = 8$

$A(3^{rd}\,\square) = 4$

$A(4^{rd}\,\square) = 2$

<u>2nd</u>:



$h = \sqrt{2^2+2^2} = \sqrt{8}$ ✓

$=$ side of 2nd square

<u>3rd</u>:

$\boxed{h^2 = 4}$

$$\text{Req Area} = \underbrace{16 + 8 + 4 + 2 + \cdots}_{\text{GP } (\infty \text{ series})}$$

$$r = \frac{8}{16} = \frac{1}{2} < 1$$

$$\therefore \quad S_\infty = \frac{a}{1-r} = \frac{16}{1-\frac{1}{2}}$$

$$= 32\,\text{cm}^2$$

The perimeters of a circle, a square and an equilateral triangle are equal. Which one of the following statements is true?

(a) The circle has the largest area

(b) The square has the largest area

(c) The equilateral triangle has the largest area

(d) All the three shapes have the same area

[ME, 2018, 1 Mark (Set-2)]

1 $P_c = P_s = P_\Delta$

$\boxed{\pi D = 4a} = 3s$

$a = \frac{\pi D}{4}$

$\pi D = 3s$

$s = \frac{\pi D}{3}$

(2) $A(\circ) = \pi r^2 = \pi\left[\frac{D}{2}\right]^2 = \frac{\pi}{4} D^2 = 0.78\, D^2$

$A(\square) = a^2 = \left(\frac{\pi D}{4}\right)^2 = \frac{\pi^2}{16} D^2 = 0.61\, D^2$

$A(\triangle) = \frac{\sqrt{3}}{4} s^2 = \frac{\sqrt{3}}{4}\left(\frac{\pi D}{3}\right)^2 = \frac{\sqrt{3}\,\pi^2}{36} D^2$

$= 0.47\, D^2$